

Breast Cancer Diagnosis: Benign vs Malignant using Logistic Regression Using the Wisconsin Diagnostic Dataset

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Abstract

This project presents a binary classification approach for breast cancer diagnosis using Logistic Regression on the Breast Cancer Wisconsin (Diagnostic) dataset. The workflow includes exploratory data analysis, feature standardization, multicollinearity reduction, baseline comparison using a Dummy Classifier, and class-balanced Logistic Regression modeling. The final model achieved approximately 97% accuracy and an ROC-AUC score of 0.99, significantly outperforming the baseline model. The results demonstrate that Logistic Regression can provide accurate and interpretable performance for breast cancer classification.

Keywords— Breast Cancer, Logistic Regression, Binary Classification, Machine Learning, Exploratory Data Analysis (EDA), StandardScaler, Dummy Classifier, ROC-AUC, Confusion Matrix, Precision-Recall Curve, Feature Selection, Class Imbalance, L2 Regularization, Medical Diagnosis

1. Introduction

Breast cancer is one of the most common forms of cancer worldwide, and early diagnosis plays a critical role in improving treatment outcomes and patient survival. Machine learning techniques can support medical diagnosis by identifying patterns in clinical data and assisting in tumor classification.

For this AML-BASIC project, the Breast Cancer Wisconsin (Diagnostic) dataset was selected from the official Suggested Datasets document. The dataset contains 569 samples, each described by 30 numerical features extracted from digitized images of Fine Needle Aspiration (FNA) of breast masses. These features characterize cell nuclei properties such as radius, texture, perimeter, area, smoothness, compactness, concavity, and symmetry.

The objective of this project is to develop a binary classification model capable of distinguishing between benign and malignant tumors using Logistic Regression.

2. Project Objectives

This project focuses on building a complete and interpretable machine learning pipeline while addressing the following tasks:

- Exploratory Data Analysis (EDA)
- Feature preprocessing and selection
- Handling class imbalance
- Baseline model comparison
- Logistic Regression modeling
- Performance evaluation and interpretation
- Visualization and discussion of model behavior

The project emphasizes both predictive performance and interpretability, which are particularly important in medical applications.

3. Hypothesis

The morphological characteristics of breast cell nuclei contain sufficient discriminative information to distinguish between benign and malignant tumors. Logistic Regression can learn meaningful relationships between these features and tumor diagnosis, allowing accurate binary classification.

4. Dataset Description

The Breast Cancer Wisconsin (Diagnostic) dataset contains:

- **569** observations
- **30** numerical features
- 1 target variable (diagnosis)

The diagnosis variable contains two classes:

- **Benign (B)** → encoded as 0
- **Malignant (M)** → encoded as 1

The features are grouped into:

- Mean values
- Standard errors (SE)
- Worst values

Examples of features include:

- radius_mean
- texture_mean
- perimeter_mean
- area_mean
- smoothness_mean

5. Data Loading and Preprocessing

The dataset was loaded using pandas and column names were manually assigned because the original dataset file does not contain headers.

```
import pandas as pd

column_names = [ ... ]

df = pd.read_csv('wdbc.data', header=None, names=column_names)
```

The diagnosis labels were converted into numerical values:

```
df['diagnosis'] = df['diagnosis'].map({'M': 1, 'B': 0})
```

This conversion is necessary because machine learning algorithms require numerical target values.

6. Exploratory Data Analysis (EDA)

6.1 Missing Values and Data Types

The dataset contains no missing values. All feature columns are numerical and suitable for machine learning.

Interpretation

The absence of missing values simplifies preprocessing and ensures that no imputation strategy is required.

6.2 Descriptive Statistics

Descriptive statistics revealed substantial differences in feature scales.

For example:

- radius mean reaches values above 28
- smoothness mean remains below 0.17

Interpretation

These differences indicate the need for feature standardization because Logistic Regression is sensitive to feature magnitude.

6.3 Feature Distribution Visualization

Histograms were used to visualize feature distributions.

Example Findings

- radius_mean shows slight right skewness
- smoothness mean is approximately normally distributed
- some features contain larger outlier values

Interpretation

The variation in distributions confirms that the dataset contains heterogeneous feature scales and supports the use of standardization before model training.

6.4 Class Distribution

The dataset contains:

- 357 benign samples
- 212 malignant samples

Interpretation

The dataset is moderately imbalanced, meaning the model could become biased toward the majority class. To address this issue:

- stratified train-test splitting was applied
- Logistic Regression used `class_weight='balanced'`

6.5 Correlation Heatmap

A correlation heatmap was generated to examine relationships between features.

Strong positive correlations were observed among features such as:

- Radius mean
- Perimeter mean
- Area mean

Interpretation

These features describe similar physical properties and therefore contain redundant information. High multicollinearity can negatively affect model stability and coefficient interpretation.

6.6 Correlation with Diagnosis

The strongest correlations with malignancy were:

- concave_points_worst
- perimeter_worst
- radius_worst

Interpretation

Tumor size and contour irregularity appear strongly associated with malignant tumors.

7. Feature Selection and Data Preparation

Highly correlated variables (correlation > 0.9) were removed to reduce multicollinearity.

Examples of removed features include:

- perimeter_mean
- area_mean
- radius_worst
- perimeter_worst

Interpretation

Removing redundant variables improves:

- model stability
- coefficient interpretability
- generalization capability

7.1 Train-Test Split

The dataset was divided into:

- 75% training data
- 25% testing data

Stratified sampling was used to preserve class distribution.

```
X_train, X_test, y_train, y_test = train_test_split(
    X_filtered,
    y,
    test_size=0.25,
    random_state=42,
    stratify=y
)
```

7.2 Feature Standardization

Feature standardization was applied using StandardScaler.

```
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

Interpretation

Standardization ensures that all features contribute comparably during optimization and prevents large-scale features from dominating the model.

8. Baseline Model

A Dummy Classifier was used as a baseline model.

```
dummy = DummyClassifier(strategy='most_frequent')
```

The baseline model predicts only the majority class.

Results

- Accuracy $\approx$ 63%
- Recall for malignant tumors = 0

Interpretation

Although the baseline accuracy appears moderate, the model completely fails to identify malignant tumors. This demonstrates that accuracy alone is insufficient for evaluating imbalanced medical datasets.

9. Logistic Regression Model

The final model uses Logistic Regression with:

- class_weight='balanced'
- max_iter=1000
- L2 regularization

```
model = LogisticRegression(  
    class_weight='balanced',  
    max_iter=1000  
)
```

Why Logistic Regression?

Logistic Regression was selected because:

- it is interpretable
- suitable for binary classification
- computationally efficient
- robust on structured tabular datasets

10. Model Evaluation

10.1 Classification Report

The model achieved:

- Accuracy $\approx 97\%$
- High precision and recall for both classes
- Strong F1-score

Interpretation of Metrics

Accuracy

Accuracy measures the proportion of correctly classified samples. However, in medical datasets accuracy alone can be misleading because class imbalance may hide poor minority-class detection.

Precision

Precision measures how many tumors predicted as malignant are truly malignant.

Recall

Recall measures how many actual malignant tumors are correctly identified.

In medical diagnosis, recall is particularly important because false negatives may delay treatment.

F1-score

The F1-score balances precision and recall, making it especially useful for imbalanced datasets.

10.2 Confusion Matrix

The confusion matrix shows:

- very few false positives
- very few false negatives

Interpretation

The model successfully identifies most malignant tumors while maintaining a low false positive rate. This is important because minimizing missed malignant cases is critical in clinical diagnosis.

10.3 ROC Curve and AUC

The ROC curve evaluates classification performance across different thresholds.

Results

- $AUC \approx 0.99$

Interpretation

An AUC close to 1.0 indicates excellent class separability and strong diagnostic capability.

10.4 Precision-Recall Curve

A Precision-Recall curve was included because the dataset is imbalanced.

Interpretation

The model maintains high precision and recall simultaneously, demonstrating strong performance in detecting malignant tumors while limiting false alarms.

11. Model Interpretation

11.1 Logistic Regression Coefficients

The Logistic Regression coefficients were analyzed to identify the most influential features.

Findings

Features related to:

- tumor size
- concavity
- contour irregularity

received the strongest positive coefficients.

Interpretation

Positive coefficients increase the probability of predicting malignancy, while negative coefficients contribute toward benign classification.

This demonstrates that the model learns medically meaningful relationships between cellular morphology and tumor diagnosis.

11.2 Model Behavior

The Logistic Regression model creates a linear decision boundary using weighted combinations of standardized features.

Class balancing improves minority-class detection, while L2 regularization reduces overfitting and stabilizes coefficients.

12. Discussion

The results demonstrate that relatively simple machine learning models can achieve strong performance on structured medical datasets when appropriate preprocessing and evaluation techniques are applied.

The combination of:

- feature standardization
- correlation filtering
- class balancing
- regularized Logistic Regression

resulted in strong predictive performance and good generalization.

The model substantially outperformed the Dummy Baseline, confirming that it successfully learned meaningful diagnostic patterns rather than relying on class frequency.

13. Limitations

Although the model achieved excellent performance, several limitations remain:

- The dataset is relatively small
- Data originates from a single source
- Logistic Regression assumes approximately linear decision boundaries
- More complex nonlinear relationships may not be fully captured

Future work may involve:

- cross-validation
- PCA
- ensemble methods
- nonlinear classifiers
- advanced feature engineering

14. Conclusion

This project demonstrated that Logistic Regression can effectively classify breast tumors using morphological features extracted from Fine Needle Aspiration biopsies.

After preprocessing, feature filtering, standardization, and class balancing, the model achieved approximately:

- 97% accuracy
- AUC close to 1.0

The model significantly outperformed the Dummy Baseline and demonstrated strong capability in identifying malignant tumors.

The results show that interpretable machine learning models can provide highly effective performance on structured medical datasets while remaining suitable for clinical interpretation.

References

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